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RESEARCH-ARTICLE

SysMix: Mixed-Size Placement for Systolic-Array-Based Hierarchical Designs

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ABSTRACT

Systolic array designs are gaining popularity due to their applications in hardware acceleration for ML computing, such as CNNs and transformers. Increasingly large ML models necessitate very high circuit energy-efficiency, which is highly correlated with minimizing placement wirelength in chip physical design. However, existing placement techniques are mostly general purpose and overlook unique properties of systolic array designs. We propose a mixed-size placement approach, called SysMix, which is tailored for systolic arrays and leverage their partial regularity in hierarchical design methodologies. Experimental results from multiple CNN designs show that SysMix achieves 53% wirelength reduction and 15X speedup compared to a commercial placer and a state-of-the-art academic placer.

KEYWORDS

Mixed-size placement, systolic array, regularity

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1 INTRODUCTION

Systolic array, composed of an array of PEs (Processing Elements), serves as a cornerstone for massively parallel computing. Systolic arrays are popular in hardware acceleration for ML (Machine Learning) computing, such as CNNs (Convolutional Neural Networks) and transformers, and have been adopted in industry, e.g., Google TPU [11]. As shown in Figure 1, systolic arrays are characterized by their highly regular structure, coupled with interconnect dominated by local connections, i.e., connections between adjacent PEs as opposed to non-adjacent PEs. Intuitively, such a regular structure is very friendly to layout designs and can potentially facilitate improved layout solution quality. Since all PEs in a systolic array are

identical, they can be pre-designed as hard IP blocks, a.k.a. macros. Such macro IP-based design reuse is commonly seen in hierarchical design methodologies and SoC designs. At the same time, the control logic circuit does not exhibit such a regular structure and is distributed throughout the array. Therefore, it can be regarded as a set of dust cells. We will present a mixed-size placement approach for such systolic array designs in hierarchical design methodologies.

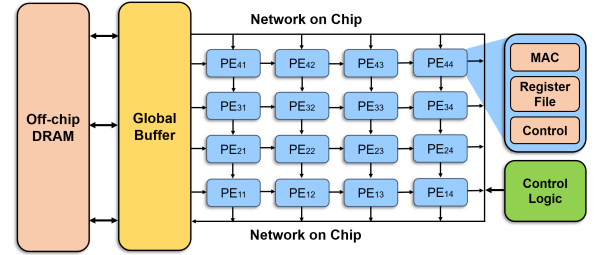


Figure 1: A systolic array-based CNN circuit architecture.

To the best of our knowledge, our work is the first study on mixed-size placement tailored for systolic array designs on ASICs (Application-Specific Integrated Circuits) by leveraging their regularity. Although mixed-size placement has been extensively studied in the past [3, 6, 15, 1, 4], they are mostly general-purpose and pay almost no attention to design regularities. Design regularity is considered in datapath placement [7, 20]. In addition to the difference between systolic array's 2D regularity and conventional datapath's 1D regularity, dust cells exist between PEs of a systolic array while datapath cells form contiguous rows without random logic cells in the middle. Therefore, datapath placement techniques are not directly applicable to systolic array designs. A macro placement approach for TPU designs is introduced in [19]. It is a reinforcement learning approach with little emphasis on exploiting the design regularity. Systolic array placement is also studied for FPGAs (Field-Programmable Gate Arrays) [14, 10].

Although the work of [9] is for placement of systolic array designs, it treats all PE cells as dust cells and solves a cell placement problem instead of mixed-size placement. Thus, it is for flat designs as opposed to hierarchical designs as in our work. Both flat design and hierarchical design are popular methodologies, and each has its own advantages and disadvantages. In a hierarchical design methodology, where the PE design is purchased from an IP vendor, it makes little business sense to flatten the PE design and use cell placement like [9]. Likewise, in a flat design methodology, it is usually better to apply cell placement like [9] than macro-based mixed-size placement like our work. Therefore, the work of [9] and



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ours are designed for two different application scenarios and not directly comparable.

The exploitation of structural regularity in placement of systolic array designs is not as straightforward as it appears to be. Due to the PE connections with global buffer, control logic cells, and IO pins, placing PEs in a strictly regular array usually results in increased wirelength. Instead, small wirelength results often exhibit partial regularity, where the spatial orders among PEs generally follow the schematic but there is not necessarily alignment for rows and columns. To solve the problem of mixed-size placement with partial regularity, we propose SysMix, an approach with three key ingredients: (1) PE-driven dust cell clustering; (2) regularity-aware macro/PE placement via LP (Linear Programming); (3) dust cell placement with a new initial placement technique. In SysMix, the regularity is exploited to reduce solution search space and help achieve better solution quality. The benefits of exploiting the regularity are clearly observed in the experimental results. The contributions of this work are summarized as follows.

- This is the first study on exploiting regularity for mixed-size placement of systolic arrays in hierarchical designs, to the best of our knowledge.
- A dust cell clustering technique is proposed for reserving the right amount of space in PE placement.
- A linear programming-based macro placement technique that makes use of partial regularity is developed.
- A dust cell placement technique with customized initial placement is proposed.
- SysMix is compared with 5 other methods on multiple CNN designs. Only SysMix, a commercial tool and another state-of-the-art academic placer successfully completed all cases. SysMix achieves 53% wirelength reduction and 15 $\times$ speedup compared to the commercial tool and the academic placer.

2 PREVIOUS WORKS

Mixed-size Placement. NTUplace3 [3] is an early influential work on mixed-size placement based on multi-level analytical techniques. RePlace [6] is an electrostatics-based analytical placer, which outperforms most of its prior works on public domain benchmark suites and forms the key basis for many recent placers. DREAMPlace [15] is a very fast analytical placer leveraging neural network training infrastructure and GPUs. AutoDMP [1] is a mixed-size placement framework based on DREAMPlace. It uses multi-objective Bayesian optimization to fine-tune the parameters of the DREAMPlace engine, and achieves PPA (Performance, Power, Area) comparable to commercial tools. The work of [4] proposes a stronger mixed-size global placement backbone based on DREAMPlace. It archives better robustness for convergence, and better HPWL (Half-Perimeter Wire-Length) and PPA.

Macro Placement. Macro placement has been extensively studied, and we summarize some recent research results in this area. MDP-trees [16] is a simulated annealing-based macro placement approach, where the MDP (Multi-Domain Packing) tree is a new solution representation technique. A deep reinforcement learning-based macro placement approach is presented in [19] and has been applied in Google TPU designs. RTL-MP [12] is a simulated annealing-based

macro placer that exploits RTL information regarding logical hierarchy. It achieves equally good or better PPA than a commercial tool and human experts, but suffers from scalability in runtime. A comparative study on different macro placement and mixed-size placement methods is provided in [5]. The results show that routed wirelength, area, and power are often highly correlated. AutoDMP and a commercial tool often achieve the best results in this study.

Datapath Placement. In [7], datapath cells are first clustered into a single macro and then a mixed-size placement is performed. Next, the datapath macro is declustered and datapath cells are placed while the random logic cells are fixed. It concludes with a random logic placement fine-tuning stage. The work of [20] simultaneously places datapath cells and random logic cells through a force-directed algorithm. Pseudo nets are inserted to align cells across multiple-bit slices to address datapath regularity.

Systolic Array Design Placement. In [14], a simulated annealing-based FPGA placement technique is developed considering the 2D regularity of systolic arrays. It achieves a 2X to 28X speedup compared to VPR [17], a well-known FPGA layout tool, with limited circuit performance loss. The work of [9] proposes a global placement framework using soft 2D regularity constraints for systolic array-based CNN designs. It places PE cells instead of PE macros in flat design methodologies. Compared with RePlace, it reduces HPWL by 6%. Since flat designs have fewer constraints than hierarchical designs, they may lead to better solution quality at the expense of longer turn-around time. Although it is also a systolic array placer, comparing a mixed-size placement and a cell placement of the same circuit would mostly disclose the difference between a hierarchical design methodology versus a flat design methodology instead of the effectiveness of placement algorithms. A new algorithmic approach for systolic array placement on FPGAs is proposed in [10]. Compared with other FPGA placement tools, it obtains a 24% reduction in HPWL.

Summary. There is almost no prior study on mixed-size placement for systolic array designs in ASICs by leveraging their regularity.

3 BACKGROUND

3.1 Graph Clustering by Bubble-FOS/C

We make use of Bubble-FOS/C [18], a graph partitioning/clustering algorithm, in our mixed-size placement. It is a connectivity-driven approach rather than minimizing cut sizes. Its framework has three stages: (1) randomly choosing a certain number of nodes as seeds; (2) growing each cluster from a seed node to its neighboring nodes by simulating fluid diffusion till all nodes are clustered; (3) for each cluster, finding its center as the new seed and regrowing from the new seeds to form new clusters. In our work, only stage 2 is needed as the seeds are PEs, which do not need to be randomly selected at the beginning or reselected after stage 2. Stage 2 is an iterative process. In each iteration, a certain amount of fluid moves from node v_j to another v_i with probability $A_{i,j}$ if there is an edge (v_i, v_j) in the given graph. The probability depends on the connectivity between v_i and v_j . At the end, a node is assigned to a cluster if its fluid mostly comes from the seed of this cluster. This iterative simulation of diffusion behaves like a Markov chain, and

probabilities $A_{i,j}$ can be treated as state transition probabilities of the Markov chain.

3.2 Background on RePlace

The dust cell placement of SysMix is based on RePlace [6], which is briefly reviewed here. Let $\lambda = (x, y)^T$ denote the vector of the coordinates of dust cells and macros. The objective is:

$$\min_{\lambda} W(\lambda) + \beta \cdot D(\lambda) \quad (1)$$

where $W(\lambda)$ is the cost function of wirelength, $D(\lambda)$ denotes the cost of cell density, and β is a weight factor. RePlace uses HPWL as the wirelength function. The HPWL of a net is approximated by the WA (Weighted-Average) model. The x component of the WA wirelength is:

$$\tilde{W}_x(\text{net}) = \frac{\sum_{i \in \text{net}} x_i e^{x_i/\gamma}}{\sum_{i \in \text{net}} e^{x_i/\gamma}} - \frac{\sum_{i \in \text{net}} x_i e^{-x_i/\gamma}}{\sum_{i \in \text{net}} e^{-x_i/\gamma}} \quad (2)$$

where γ is a smoothing factor. The density function $D(\lambda)$ is used to reduce overlap among cells. RePlace models each cell as electric charges proportional to its area. The repelling forces among the charges naturally spread out cells during placement. For a placement canvas evenly divided into an $B \times B$ bin grid, the density function $D(\lambda)$ is modeled as the total potential energy:

$$\Phi(\tilde{x}, \tilde{y}) = \sum_{0 \leq \tilde{x} < B} \sum_{0 \leq \tilde{y} < B} \rho(\tilde{x}, \tilde{y}) \psi(\tilde{x}, \tilde{y}) \quad (3)$$

where $\tilde{x}/\tilde{y}$ denotes the column/row indices of bins, ρ denotes the charge density of a bin, and ψ represents the local electric potential.

RePlace solves the placement problem with unconstrained optimization techniques. However, as stated in [4], the mixed sizes of cells often cause issues with optimization convergence. Therefore, we mainly adopt it for the dust cell placement part of our method.

4 PROBLEM FORMULATION

The input is a systolic array-based design characterized by a hypergraph $G = (V, E)$. The node set V is composed of a set of PEs P , a set of dust cells C and a set of fixed pins T corresponding to IOs and preplaced global buffer, i.e., $V = P \cup C \cup T$. Set P is an $M \times K$ array of PEs, where M and K are natural numbers. The hyperedge set E is the netlist indicating connections among PEs, dust cells, and fixed pins. Each PE is treated as a hard macro, and its internal nets are neglected. The mixed-size placement is to decide PE and dust cell locations in a 2D plane such that the total HPWL (Half Perimeter Wire-Length) is minimized and there is no overlap among any PEs, dust cells, and fixed pins. Mixed-size placement for systolic array designs exploiting regularity is a relatively new problem. As such, it will be tackled in a step-by-step manner with an emphasis on HPWL in this work like many other mixed-size placement works. Other objectives, such as routability, will be addressed in our future research.

5 THE PROPOSED SYSMIX PLACEMENT

5.1 Overview

Although the structure of a systolic array is very regular, its placement is not straightforward for two reasons. First, there are wire connections between PEs and fixed pins, which may distort the

regularity. Second, there are a significant number of dust cells interleaved between PEs, which also disturb the regularity. The effect of fixed pins and dust cells on regularity is demonstrated in [9]. Such partial regularity is significantly more difficult to exploit than complete regularity. A similar difficulty is also observed in datapath placement [20]. On the other hand, even partial regularity provides an opportunity for further HPWL reduction.

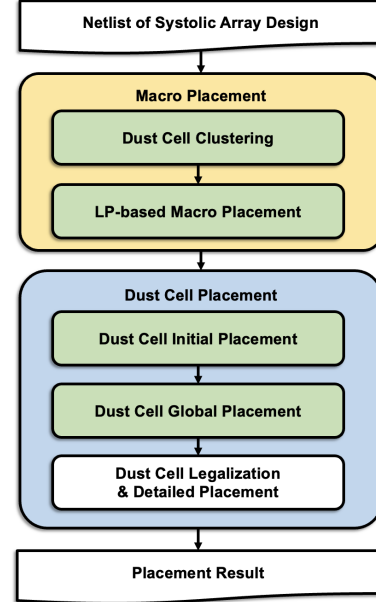


Figure 2: Overview of the proposed SysMix placement method, where the green parts are our contributions.

An overview of our proposed SysMix placement methodology is provided in Figure 2. In the phase of macro placement, PEs are placed with the consideration of dust cells through dust cell clustering. The macro placement is realized through LP. In the second phase, dust cells are placed while the PEs are fixed. Both the macro placement and dust cell placement leverage the partial regularity of systolic array designs.

5.2 Macro Placement

5.2.1 Dust Cell Clustering for C3-Macro Generation. A **C3-macro** consists of a PE surrounded by a subset of related dust cells, like a Chocolate-Covered Cherry. The PE is a hard macro (cherry) and its surrounding dust cells form a firm shell (chocolate). In this step, every dust cell is clustered to a certain PE to form an $M \times K$ array of C3-macros. Such C3-macros provide a rough estimate of how much space to be reserved for dust cells around each PE when deciding PE locations.

This is a special graph clustering problem different from ordinary graph clustering/partitioning [2, 13]. Thus, we adopt the Bubble-FOS/C algorithm [18], which fits our need very well. The overall Bubble-FOS/C algorithm is briefly reviewed in Section 3.1. Since we can directly use PEs as cluster seeds and there is no need for further seed adjustment, only stage 2 of Bubble-FOS/C is employed here.

Starting from PEs as cluster seeds, dust cells are clustered to PEs by simulating a fluid diffusion process. As illustrated in Figure 3,

PEs are fluid sources indicated by different colors. Over the time, or simulation iterations, the fluids diffuse to neighboring dust cells according to graph connectivity. When the diffusion covers the entire graph as the rightmost part of Figure 3, all dust cells are clustered with their PEs to form an array of C3-macros.

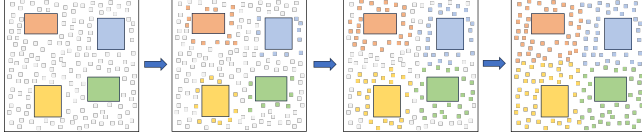


Figure 3: An illustration of clustering by simulating fluid diffusion. The four large boxes are PEs and small rectangles indicate dust cells. Fluids are represented by different colors. The fluid flow coloring result correspond to 4 clusters of dust cells.

Graph connectivity is described by edge set E of the given circuit netlist. If the degree of a net $e \in E$ is denoted by δ_e , then the connection weight for this net is defined as

$$\eta(e) = \frac{1}{2\delta_e(\delta_e - 1)}. \quad (4)$$

The diffusion simulation is based on an $N \times N$ connectivity matrix A , where N is the total number of dust cells and PEs, i.e., $N = |C| + |P|$. For two distinct nodes $v_i, v_j \in C \cup P$, the set of nets incident to both v_i and v_j is denoted by $E_{i,j} \subset E$. Their corresponding entry in connectivity matrix A is defined to be

$$A_{i,j} = \sum_{e \in E_{i,j}} \eta(e) \quad (5)$$

Since a node may be a PE, which consists of many cells, it is likely a PE $v_i \in P$ has multiple net connections with a dust cell $v_j \in C$. Each diagonal element $A_{i,i} = g$, where g is a user specified parameter. Every row of matrix A is normalized as

$$\tilde{A}_{i,j} = \frac{A_{i,j}}{\sum_{k=1}^N A_{i,k}} \quad (6)$$

After the normalization,

$$\sum_{j=1}^N \tilde{A}_{i,j} = 1, \quad i = 1, 2, \dots, N \quad (7)$$

Then, $\tilde{A}_{i,j}$ can be interpreted as the probability of fluid flowing from node v_j to v_i . Although the matrix may contain symmetry, i.e., $\tilde{A}_{i,j} = \tilde{A}_{j,i}$, fluid flows from a node with the greater amount of fluid to its neighbor node with the less amount of fluid. A diagonal entry $\tilde{A}_{i,i}$ means the probability that fluid stays at its current node. If g is too small, fluid diffuses too fast and little fluid would be left at the nodes near their sources. If g is too large, it would take much more iterations for the fluids to cover all nodes.

The fluid sources are described by an $N \times |P|$ matrix S , whose entries are defined by

$$S_{i,j} = \begin{cases} 1, & i = j + |C| \\ 0, & \text{others} \end{cases} \quad (8)$$

where each column corresponds to a PE. The upper $|C|$ rows of this matrix correspond to dust cells and are all zeros. The lower $|P|$

rows imply interactions between dust cells and PEs, and form an identity sub-matrix. For $|C| = 4$ and $|P| = 3$, the S matrix is

$$S = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

An $N \times |P|$ distribution matrix D has entries $D_{i,j}$ indicating the amount of fluid from source v_j ending up at v_i . At the beginning, $D_0 = S$ for the initial distribution, where all fluids are at their source nodes. From iteration $k-1$ to k , $D_k = \tilde{A}D_{k-1}$, and therefore

$$D_k = \tilde{A}^k S \quad (9)$$

The iterations terminate when all entries of D_f are non-zero in iteration f . For each dust cell $v_i \in C$, we find column j in D_f such that $D_{i,j}$ is the maximum in row i of matrix D_f . Then, v_i is assigned to PE $v_j \in P$ in the final clustering solution, as v_i receives more fluid from v_j than any other PEs.

A C3-macro is composed of a PE and the dust cells assigned to this PE by the clustering process. The area of a C3-macro is its PE area plus all its dust cell areas. We try to make its aspect ratio conform with its PE and as close to a square as possible. In the subsequent macro placement, all pins of a C3-macro, including those from its PE and dust cells, are assumed to be at the C3-macro center.

5.2.2 C3-Macro Placement. C3-macros are placed through LP (Linear Programming). The problem formulation is to decide the locations of all C3-macros such that total HPWL is minimized while there is no overlap between any two C3-macros. Please note that the HPWL here covers only the nets among C3-macros and fixed pins while the nets incidents to dust cells are contained within C3-macros and thus not included. The HPWL is estimated by

$$HPWL = \sum_{e \in E'} \max_{u \in e} x_u - \min_{v \in e} x_v + \max_{u \in e} y_u - \min_{v \in e} y_v \quad (10)$$

where E' indicates the nets among C3-macros and fixed pins. A C3-macro is identified by the row and column indices of its PE in the systolic array. In other words, a C3-macro $\mu_{i,j}$ is in row i and column j of the corresponding systolic array.

Due to the regularity of systolic arrays, we can enforce non-overlapping constraints for only a local neighborhood of each C3-macro instead of all C3-macro pairs. In this manner, the relevant ordering of the PEs in the schematic is followed while the center of PEs are not necessarily aligned in each row and column. There are two such local non-overlapping constraints.

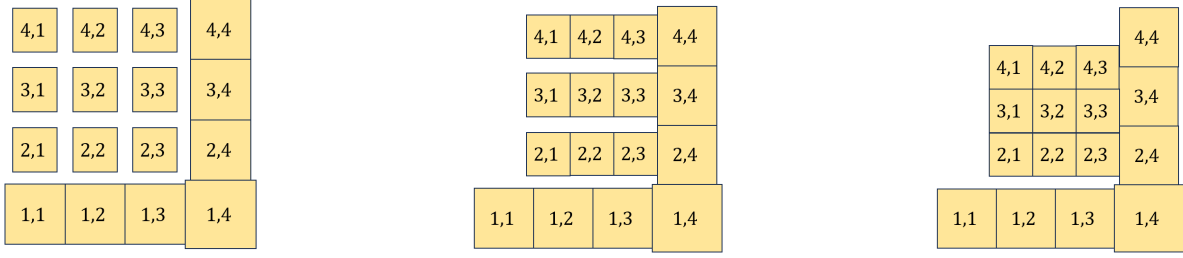
Horizontal non-overlapping constraint. For each C3-macro $\mu_{i,j}$, we require

$$x_{i,j} - x_{i-1,j-1} \geq \frac{w_{i,j} + w_{i-1,j-1}}{2} - \epsilon_x \quad (11)$$

$$x_{i,j} - x_{i,j-1} \geq \frac{w_{i,j} + w_{i,j-1}}{2} \quad (12)$$

$$x_{i,j} - x_{i+1,j-1} \geq \frac{w_{i,j} + w_{i+1,j-1}}{2} - \epsilon_x \quad (13)$$

where $w_{i,j}$ represents the width of C3-macro $\mu_{i,j}$ and $\epsilon_x \geq 0$ is a user specified parameter. Constraint (12) requires that for two


 (a) C3-macro placement when $\epsilon_x = \epsilon_y = 0$.

 (b) First pass LP when $\epsilon_y = 0$ and $\epsilon_x > 0$.

 (c) Second pass LP when $\epsilon_y > 0$.

 Figure 4: C3-macro placement with non-overlapping constraints of different ϵ_x and ϵ_y settings.

C3-macros $\mu_{i,j}$ and $\mu_{i,j-1}$, which are in the same row, $\mu_{i,j}$ must be strictly on the right of $\mu_{i,j-1}$. However, for diagonal neighbors in adjacent rows, e.g., $\mu_{i,j}$ and $\mu_{i+1,j-1}$, horizontal overlap up to ϵ_x is allowed according to constraints (11) and (13). For example, in Figure 4(b), $\mu_{1,2}$ and its diagonal neighbor $\mu_{2,1}$ have horizontal overlap of $\Delta x \leq \epsilon_x$. This limited relaxation allows more compact placement, which can be seen by comparing Figures 4(a) and (b).

Vertical non-overlapping constraint. For each C3-macro $\mu_{i,j}$, we require that

$$y_{i,j} - y_{i-1,j-1} \geq \frac{h_{i,j} + h_{i-1,j-1}}{2} - \epsilon_y \quad (14)$$

$$y_{i,j} - y_{i-1,j} \geq \frac{h_{i,j} + h_{i-1,j}}{2} \quad (15)$$

$$y_{i,j} - y_{i-1,j+1} \geq \frac{h_{i,j} + h_{i-1,j+1}}{2} - \epsilon_y \quad (16)$$

where $h_{i,j}$ represents the height of C3-macro $\mu_{i,j}$ and $\epsilon_y \geq 0$ is a user specified parameter. Large values of ϵ_x and ϵ_y allow more compact placement with risk of not being legalizable. Moreover, $\epsilon_x > 0$ and $\epsilon_y > 0$ cannot simultaneously hold, otherwise overlap between two diagonally neighboring C3-macros may appear. In other words, either $\epsilon_x > 0$ or $\epsilon_y > 0$. Such either-or constraints can be handled by ILP (Integer Linear Programming) but not LP. However, ILP is known to be slow and have poor scalability. To solve this difficulty, we propose a 2-pass LP approach, where only one of $\epsilon_x > 0$ and $\epsilon_y > 0$ is true in each pass.

2-pass LP methodology. Due to different C3-macro sizes, unnecessary spacing may exist among small C3-macros if $\epsilon_x = \epsilon_y = 0$. This is illustrated in Figure 4(a). To avoid such spacing and thereby reduce HPWL, we propose a 2-pass LP approach, where one LP is performed after another, and the objective function (minimizing HPWL) and non-overlapping constraints are almost the same except the ϵ values.

In the first pass LP, we set $\epsilon_y = 0$ and $\epsilon_x > 0$. Then, C3-macros are horizontally compacted as shown in Figure 4(b). In the second pass LP, if a C3-macro does not have horizontal overlap with its diagonal neighbors from the first pass, their corresponding ϵ_y is set to be greater than 0. For example, in Figure 4(b), $\mu_{4,3}$ and $\mu_{3,4}$ do not have horizontal overlap. Hence, their corresponding ϵ_y is greater than 0 in the second pass LP and vertical overlap of Δy between them is allowed as shown in Figure 4(c). Comparing Figure 4(c) and (a), our 2-pass LP allows much more compact placement without incurring overlap among C3-macros.

5.3 Dust Cell Placement

In this phase, dust cells are declustered from C3-macros and placed while the PE locations are fixed. Our dust cell placement includes three steps: (1) initial placement; (2) global placement; (3) detailed placement and legalization. The initial placement in our SysMix is a new technique that makes use of the regularity of systolic arrays and will be elaborated later in this section. The global placement is carried out using RePLAce [6] with a change. Although PE areas intuitively need to be counted toward cell densities in global placement, their large sizes overshadow the density of dust cells, which are to be placed in this phase. In order to let RePLAce focus on dust cells instead of PEs, we scale PE dimensions with a factor the same as the target density throughout our execution of RePLAce-based dust cell placement. Step 3, detailed placement and legalization, is realized by calling NTUPlace3 [3].

In conventional cell placement methods [3, 6], initial solutions are generated randomly or through quadratic placement. In our SysMix, we propose a new initial solution technique for dust cells by leveraging the regularity of systolic array designs. The main idea is to use the PE placement as a backbone and then place dust cells using the results of clustering described in Section 5.2.1.

The initial placement is based on dust cell connections with PEs. For simplicity, we use cells to refer dust cells. The netlist connections are described by matrix A , which is described in Section 5.2.1. It is an $N \times N$ matrix, where its upper-left $|C| \times |C|$ sub-matrix describes connections among cells, its lower-right $|P| \times |P|$ sub-matrix is for connections among PEs, the upper-right $|C| \times |P|$ sub-matrix and lower-left $|P| \times |C|$ sub-matrix indicate connections between cells and PEs. We derive a $|C| \times |P|$ matrix Φ from A to capture direct and indirect connections between a cell and a PE. There is a direct connection between a cell $v_i \in C$ and a PE $v_j \in P$ if there is an edge $e \in E$ such that $v_i \in e$ and $v_j \in e$. There is an indirect connection between a cell $v_i \in C$ and a PE $v_j \in P$ if there is another cell $v_k \in C$ such that $\{v_i, v_k\} \subseteq e \in E$ and v_k is assigned to C3-macro corresponding to v_j in the clustering described in Section 5.2.1.

The process for generating the cell-PE connection matrix Φ is outlined in Algorithm 1. Its lines 4 and 5 are to check the upper-left sub-matrix of A to identify indirect connections, which are counted in matrix Φ in lines 6 and 7. Lines 10-14 examine the upper-right sub-matrix of A to account for the direct connections in Φ . After the normalization in line 16, $\Phi_{i,j}$ can be interpreted as the normalized connection weight between dust cell $v_i \in C$ and PE $v_j \in P$.

Let vectors $(\mathbf{x}_P, \mathbf{y}_P)$ denote the fixed PE locations, then our initial dust cell placement $(\mathbf{x}_C, \mathbf{y}_C)$ is obtained by Equation (17).

Algorithm 1 Generating Cell-PE Connection Weight Matrix

```

1: procedure CELLPE-CONNECTION(Connection matrix  $A$ )
2:   Initialize  $|C| \times |P|$  matrix  $\Phi$  with all 0's
3:   for  $i$  in  $1, \dots, |C| - 1$  do // for all dust cells
4:     for  $j$  in  $i + 1, \dots, |C|$  do //check connection to other cells
5:       if  $A(i, j) > 0$  then
6:          $c_j \leftarrow$  C3-macro ID of cell  $v_j$  in clustering
7:          $\Phi(i, c_j) += A(i, j)$  // indirect connect with PE
8:       end if
9:     end for
10:    for  $j$  in  $1, \dots, |P|$  do //check connection to PEs
11:      if  $A(i, j + |C|) > 0$  then
12:         $\Phi(i, j) += A(i, j)$  // direct connect with PE
13:      end if
14:    end for
15:  end for
16:  Normalize( $\Phi$ ) such that  $\sum_{j=1}^{|P|} \Phi_{i,j} = 1, i = 1, \dots, |C|$ 
17:  return  $\Phi$ 
18: end procedure

```

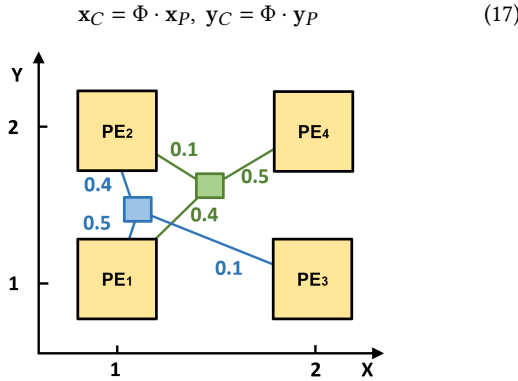


Figure 5: An illustration of dust cell initial placement. The small boxes indicate dust cells. The numbers along wire segments are non-zero elements of connection weight matrix Φ .

This is illustrated by an example in Figure 5, where

$$\Phi = \begin{bmatrix} 0.5 & 0.4 & 0.1 & 0.0 \\ 0.4 & 0.1 & 0.0 & 0.5 \end{bmatrix}, \mathbf{x}_P = [1, 1, 2, 2]^T, \mathbf{y}_P = [1, 2, 1, 2]^T$$

Then, the initial placement of the two dust cells is obtained by

$$\mathbf{x}_C = \Phi \cdot \mathbf{x}_P = \begin{bmatrix} 1.1 \\ 1.5 \end{bmatrix}, \mathbf{y}_C = \Phi \cdot \mathbf{y}_P = \begin{bmatrix} 1.4 \\ 1.6 \end{bmatrix}$$

One can see that the computations of Algorithm 1 and Equation (17) have very limited complexity. Indeed, it is observed that our initial placement achieves better results than quadratic placement with much shorter runtime. It also enables significantly shorter HPWL compared to random initial placement.

6 EXPERIMENTAL RESULTS

6.1 Experiment Setup

Several systolic array-based CNN circuit designs are obtained according to [21] for ASIC style. These circuits are synthesized using Synopsys Design Compiler with the ASAP7 cell library [8]. They

cover different dataflow types, WS (Weight Stationary) and OS (Output Stationary), and various PE array sizes. The characteristics of these circuits are summarized in the first column of Table 1.

We compared our method (**SysMix**) with the following state-of-the-art mixed-size placers

- **RePlace**[6];
- **DREAMPlace**[15];
- **AutoDMP**[1];
- The work of [4];
- A **commercial placement tool**.

AutoDMP usually generates multiple solutions with different PPA (Performance Power Area) tradeoffs. We pick its minimum HPWL solution for a fair comparison. Except AutoDMP and [4], all the other methods are evaluated on an 8-thread CPU machine with 3.4GHz frequency and 250GB RAM. AutoDMP can only be performed on a GPU platform, and it runs on NVIDIA GeForce RTX 2080 Ti in this experiment. The result of [4] was obtained from an NVIDIA GeForce RTX 2080 GPU.

6.2 Main Results

The main results of HPWL and total runtime are summarized in Table 1. All the HPWL values except those labeled with illegal are legalized results. For those results labeled with illegal, global placement was successful, and their global placement HPWL results are reported if available. Please note that each tool has its own legalizer, e.g., the legalization of DREAMPlace was performed using DREAMPlace's legalizer. The global placement of RePlace cannot converge for circuit WS8 and circuit WS64.

Among the six methods being compared, only our SysMix, the commercial tool, and AutoDMP can successfully find legalized solutions for all circuits. Compared to the commercial tool and AutoDMP, our SysMix can reduce wirelength by over 53% on average and are $15\times - 217\times$ faster. Although DREAMPlace's results look close to ours, DREAMPlace failed to legalize five out of the eight circuits. Among all methods, RePlace is the fastest, but it failed in three circuits, and the available HPWL results are 78% larger than our SysMix on average. The method of [4] failed in legalizing WS32 and generated $2.7\times$ HPWL compared to our SysMix.

6.3 Ablation Studies

6.3.1 Effectiveness of LP-Based C3-Macro Placement. We compared our SysMix with two alternative approaches, which perform the PE placement differently.

- (1) Regular PE placement. PEs instead of C3-macros are placed in a regular 2D array, where rows and columns are well aligned. Then, dust cells are placed using RePlace with the PEs are fixed. This approach is straightforward yet the tricky part is the spacing among adjacent PEs. Since this is a regular array, the spacing for different columns and rows are the same. Too small spacing may cause legalization failures while too large spacing increases HPWL unnecessarily. For each case, we conduct multiple runs and choose the minimum spacing that allows successful legalization.
- (2) RePlace with regular initial PE placement (RePlace-RI). This is almost the same as RePlace except that the initial PE placement is a regular 2D array instead of a random placement.

Table 1: Comparison of (legalized unless specified) HPWL and total runtime. For results that failed in legalization, global placement HPWL values are reported. #cells means the number of dust cells.

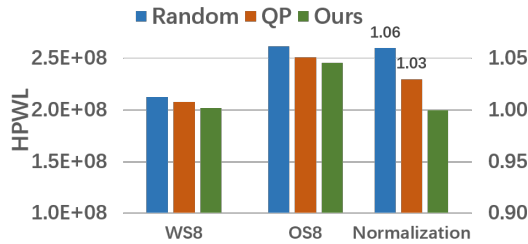
Circuit	PE array	#cells	SysMix		Commercial Tool		RePlAce		DREAMPlace		AutoDMP		Ref [4]		
			HPWL	RT(s)	HPWL	RT(s)	HPWL	RT(s)	HPWL	RT(s)	HPWL	RT(s)	HPWL	RT(s)	
WS8	8×8	5K	3.72E+08	13	4.38E+08	96	Not converge	N/A	4.97E+08	Illegal	10	5.58E+08	3,654	2.37E+09	19
OS8	8×8	9K	4.06E+08	15	5.13E+08	151	4.38E+08	6	4.97E+08	Illegal	9	6.55E+08	5,083	5.67E+08	19
WS16	16×16	17K	1.70E+09	21	1.78E+09	381	1.99E+09	15	1.53E+09	Illegal	19	3.38E+09	3,052	7.65E+09	124
OS16	16×16	33K	2.08E+09	41	2.04E+09	689	2.42E+09	24	2.28E+09	Illegal	57	4.58E+09	5,596	3.46E+09	124
WS32	32×32	105K	6.12E+09	30	3.78E+10	882	2.26E+10	21	5.62E+09	Illegal	32	1.19E+10	10,610	6.47E+09	16
OS32	32×32	172K	2.34E+10	217	5.50E+10	2175	2.47E+10	191	2.38E+10	Illegal	231	3.70E+10	13,272	3.49E+10	57
WS64	64×64	268K	5.76E+10	562	1.90E+11	8,329	Not converge	N/A	4.27E+10	Illegal	763	1.57E+11	11,527	5.36E+10	93
OS64	64×64	463K	9.32E+10	798	4.10E+11	12,135	1.19E+11	854	9.49E+10	Illegal	1052	6.63E+11	24,032	1.69E+11	93.0
Normalized average			100%	1	216%	15.22	178%	0.59	108%	0.96	180%	217	274%	1.75	

Since this is merely an initial solution, the spacing among rows and columns is not important.

Table 2: Comparisons with regular PE placement and RePlAce with regular initial PE placement.

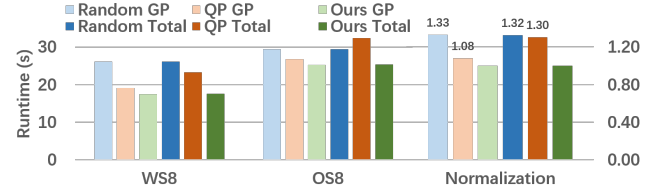
Circuit	SysMix		Regular PE Placement		RePlAce with regular initial PE placement	
	HPWL	RT(s)	HPWL	RT(s)	HPWL	RT(s)
WS8	3.72E+08	13	5.12E+08	10	Not converge	N/A
OS8	4.06E+08	15	6.24E+08	11	4.25E+08	6
WS16	1.70E+09	21	7.82E+09	21	1.94E+09	14
OS16	2.08E+09	41	3.26E+09	65	2.39E+09	22
WS32	6.12E+09	30	7.23E+09	47	6.82E+09	34
OS32	2.34E+10	217	3.89E+10	265	3.61E+10	243
Norm avg	100%	1	199%	1.15	120%	0.71

The comparison results are shown in Table 2. Compared to the regular PE placement, SysMix can reduce legalized HPWL by almost a half and decrease runtime by 15%. This shows that the impact of dust cells and IO pins is complicated and cannot be handled by a very regular PE layout. RePlAce-RI shows much better results than the RePlAce results, which are shown in Table 1. However, it still causes 20% higher HPWL, which is quite significant for placement. Moreover, RePlAce cannot yet converge for WS8, even with the regular initial solution.


Figure 6: Legalized HPWL from different dust cell initial placement methods.

6.3.2 Effectiveness of Dust Cell Initial Placement. In order to examine the effectiveness of our proposed dust cell initial placement described in Section 5.3, we compare it with random placement and Quadratic Placement (QP), two commonly used conventional initial placement techniques, while the other parts of the SysMix

method are retained unchanged. The results are summarized in Figures 6 and 7, where the HPWL values are obtained after legalization. Compared to random initial placement, our method reduces HPWL and global placement runtime by 6% and 33%, respectively. Compared to QP-based initial placement, our initial placement leads to 3% smaller HPWL and 30% reduction of the total runtime of initial placement and global placement. Since our initial solutions are computed very quickly and make dust cells well located, the remaining RePlAce iterations can be significantly reduced.


Figure 7: Global placement runtime and total (global+initial) placement runtime.

6.3.3 Effectiveness of Dust Cell Clustering. We compared our placer using the Bubble-FOS/C clustering (Section 5.2.1) with a naïve dust cell clustering method embedded in our placement framework. In this naïve clustering, a dust cell is clustered to its nearest neighbor PE according to the number of hops in the circuit graph. The comparison results are shown in Table 3. Compared to the naïve clustering, the Bubble-FOS/C-based clustering in our method leads to 30% reduction in HPWL and 25% total placement runtime reduction. One reason for this difference is that Bubble-FOS/C clustering is based on a relatively sophisticated connectivity metric of Equation (5), which reflects the effect of net degrees. In addition, diffusion is a natural way to capture this metric through multiple connecting paths between dust cells and PEs.

Table 3: Comparison between our clustering and a naïve clustering method.

Circuit	Our Bubble-FOS/C clustering			Naïve clustering		
	HPWL	RT	Clustering RT	HPWL	RT	Clustering RT
WS8	3.72E+08	13s	0.0018s	5.07E+08	17s	0.016s
OS8	4.06E+08	15s	0.0021s	6.18E+08	21s	0.017s
Ave	3.89E+08	14s	0.0020s	5.63E+08	19s	0.017s
Norm	100%	1	1	145%	1.34	8.5

6.3.4 Effect of Dust Cell Clustering Parameters. There are two parameters for the dust cell clustering described in Section 5.2.1. One is the diagonal element value g for the connectivity matrix A , and the other is the number of iterations. We evaluate the impact of these parameters on case WS8. Since the impact on total runtime is limited, we mainly focus on the legalized HPWL, which is depicted in Figure 8. One can see that too few iterations increase HPWL. When the number of iterations is sufficiently large, the middle choice of diagonal values g leads to the best HPWL. This is consistent with our expectation that g should not be too large or too small.

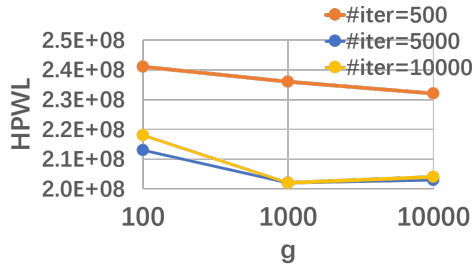


Figure 8: Effect of dust cell clustering parameters.

6.3.5 Effectiveness of PE Scaling in Dust Cell Global Placement. During the dust cell global placement in SysMix, we scale the dimensions of each PE by a factor equal to the target density such that PEs do not overshadow dust cells in bin density estimation of RePlace [6]. Despite its simplicity, this PE scaling technique plays a critical role in the success of SysMix. When PEs are not scaled as such, the global placement fails to converge for both WS8 and OS8, the two smallest cases.

7 CONCLUSIONS AND FUTURE RESEARCH

This work provides the first study on mixed-size placement for systolic array designs in hierarchical ASIC style, to the best of our knowledge. The proposed approach SysMix makes use of partial regularity in systolic array designs and produces significantly better wirelength results compared to state-of-the-art techniques including a commercial tool, which either fail in obtaining legalized solutions or lead to considerably greater wirelength. Ablation studies have been performed to demonstrate the effectiveness of each component in our SysMix placer. In future research, we will extend this work to handle routability and evaluate solutions with global routing.

8 ACKNOWLEDGMENTS

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